Jonathan Ung

Canadian Citizen | US J-1 Eligible
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Education

Simon Fraser University - Bachelor of Science in Computer Science **EXPECTED APR 2028**

Relevant Coursework: Systems Programming, Distributed Systems, Artificial Intelligence, Networking

Experience

Amazon | **Vancouver, BC, Canada** **SEP 2026 – PRESENT**

Software Engineering Intern *AWS, JavaScript, React.js, Java, KiroCrew*

- Built a diagnostic dashboard for an Amazon Business payment method that surfaces service data and configures **5 SEV-3 alarms**, enabling ops to triage failures without engineering escalation
- Designed a specialized team-specific evaluation standard for KiroCrew agent skills, building **10 scenario suites** focused around idempotency and reliability tenets

Qualcomm | **Markham, ON, Canada** **MAY 2026 – AUG 2026**

Software Engineering Intern *C/C++, Python, ONNX, QNN*

- Enabled models exceeding device RAM to run on-device by dynamically paging KV-cache partitions, allowing **100+** devices to be used for validation workflows
- Built and distributed an internal agentic developer harness used by **80+ engineers** which decomposes epics into linked JIRAs and dispatches worktree agents that open draft PRs, **saving 4+ hours/week per engineer**
- Unified **70+** NPU math and approximation functions into a single library with rigorous unit + on-device tests integrated into Jenkins CI for automated regression coverage every PR

Rivian and Volkswagen Group Technologies | **Vancouver, BC, Canada** **MAY 2025 – DEC 2025**

Software Engineering Intern *Kotlin, Java, Android, Protobuf, QNX*

- Owned **12+ user-facing experiences** and vendor integrations for Spotify, Apple Music, and Tidal on the RivianOS 2 Media App, delivering to production builds
- Migrated Media App from fragment-based architecture to MVVM with view controllers consuming data models, cutting RAM usage **40%** and accelerating feature velocity **30%**
- Led collaboration across platform and QA teams to resolve **5+** critical vehicle safety defects

SFU Robot Soccer | **Burnaby, BC, Canada** **FEB 2024 – PRESENT**

Software Team Lead *C++, Python, Qt6, Protobuf, Docker*

- Led **30-member** student development team, creating the first autonomous agent in the club's history, along with **7+** other major projects

Projects

Strike - Agentic Coding TUI *Go, WebSocket/SSE, Docker, Nix*

- Architected an **event-sourced** Go engine emitting protocol events to clients over JSONL session logs, shipping **14+ releases** across **1000+ PRs** with **4 external contributors**

Frontal-Following Robot Navigation *Python, PyTorch, ROS2, Gym*

- Trained **LSTM-based** human motion predictor achieving **90% accuracy** and integrated predicted action probabilities into **MCTS** selection for predictive frontal following behavior
- Deployed real-time navigation on **Quanser QBot** with **ZED camera** and **Vicon** localization, running planning at **5Hz**

Technical Skills

Languages: Python, C/C++, CUDA, C#, Java, Kotlin, JavaScript, TypeScript, Go, SQL, Bash
Frameworks: React, Next.js, .NET, FastAPI, Node.js, Express.js, Spring Boot, Electron, SignalR, GraphQL
Tools & Infrastructure: Docker, AWS (EC2, S3, Lambda), PostgreSQL, MongoDB, Redis, Git, CI/CD, Linux